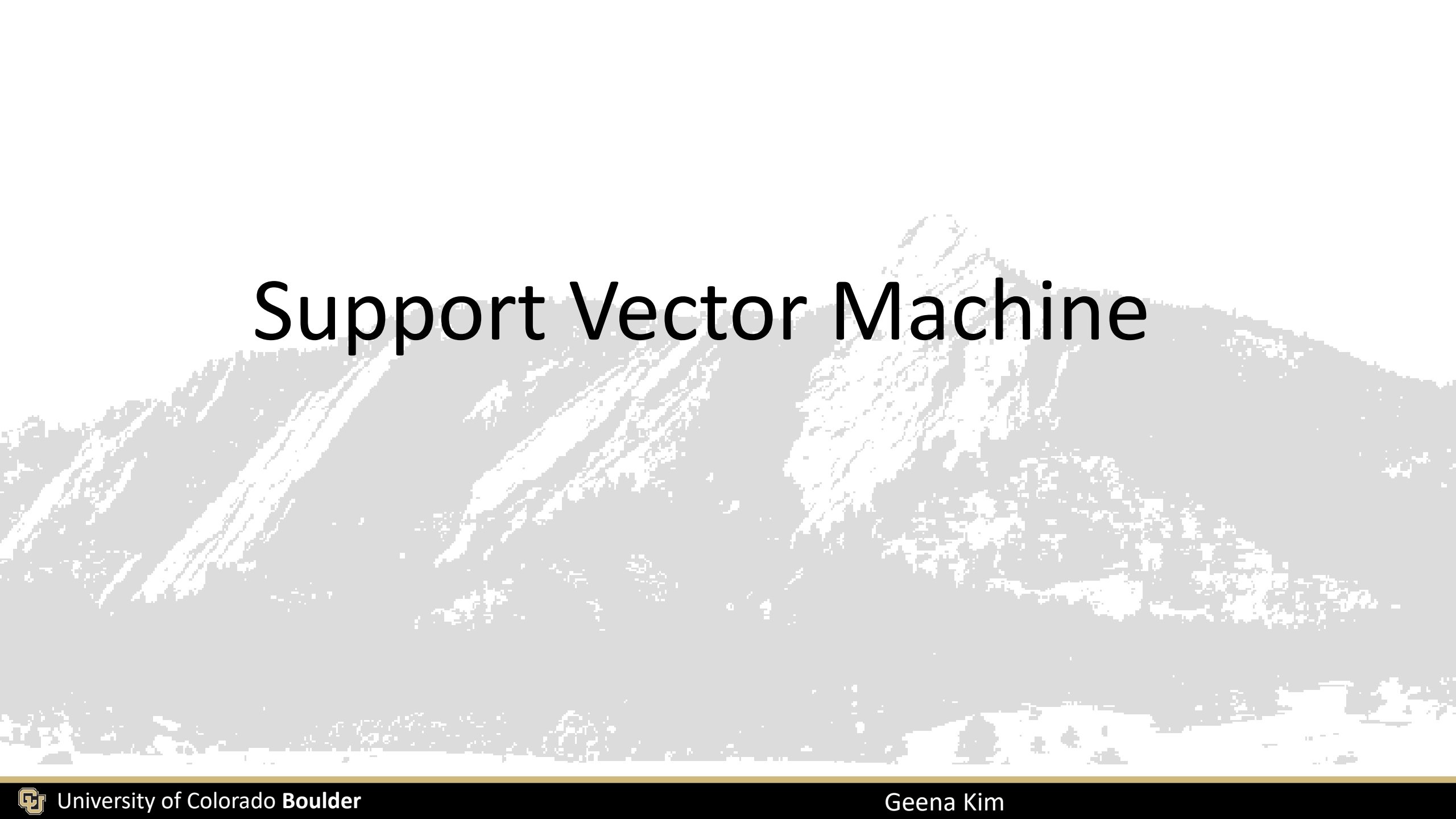
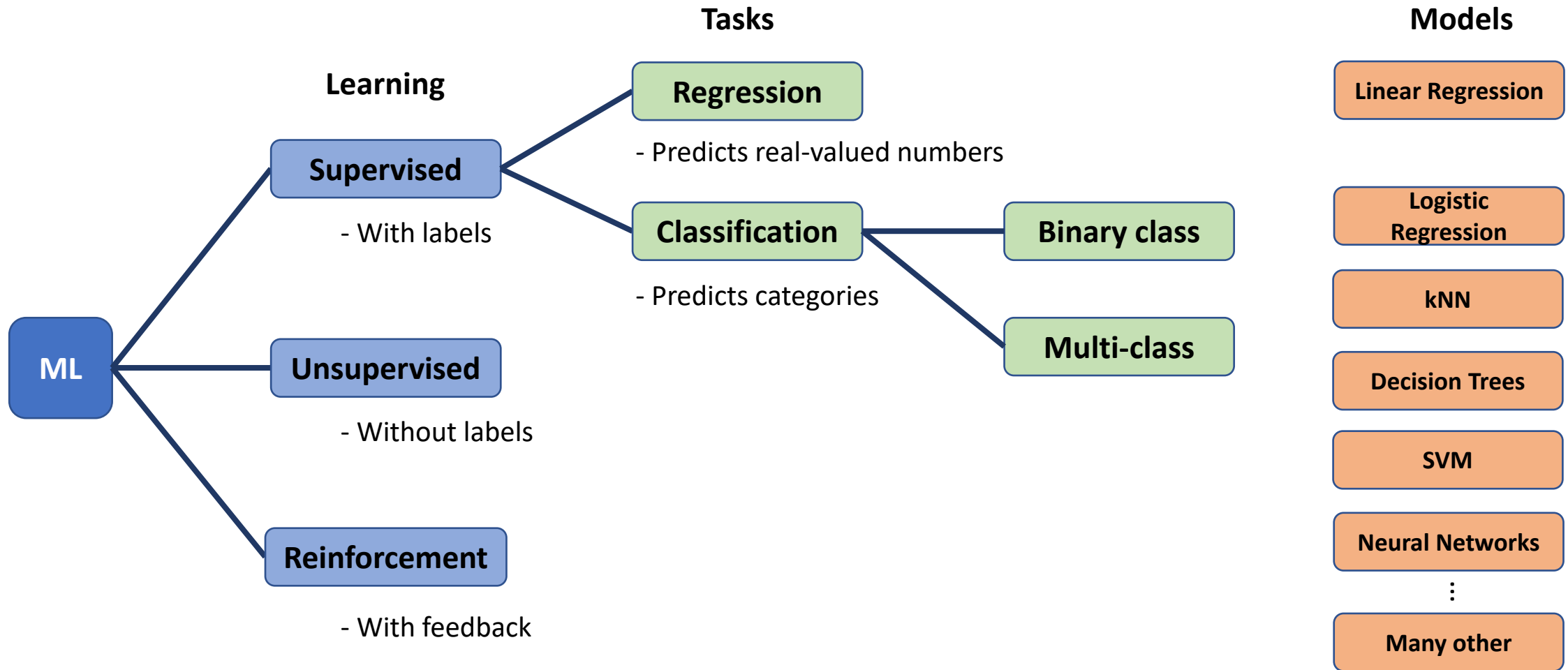


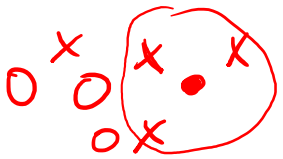
Support Vector Machine



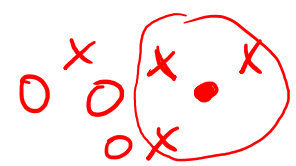
Review: Types of machine learning problems



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Models	Hyperparameters	Parameters	Loss or Criteria
Linear Regression	X	$y = w_1x_1 + w_2x_2 + w_0$	MSE / RSS
Logistic Regression	X ,	$z = \sum w_i x_i$ $\sigma(z)$	BCE
kNN	<u>k</u> , 	X	
Decision Trees			
SVM			
Neural Networks			

Review: Types of machine learning problems

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kNN	k 	X	
Decision Trees	maxdepth α #tree lr.	X X	
SVM	C	X	→
Neural Networks	✓	✓	✓

gini
entropy) classified
MSE / RSS - regression

Support Vector Machine

- Use hyperplane
- Use kernel
- Performs well
- Regression, Classification (binary class)

Review: Binary Classification

Yes or No problem

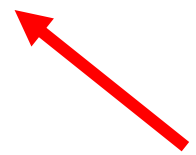
- Creditcard Default
- Fraudulent Insurance Claim
- Spam Filtering
- Medical Diagnosis
- Survival Prediction
- Customer Retention
- Image Recognition
- Sentiment analysis

Review: Logistic Function

$$P^{(i)} = \sigma(z^{(i)})$$

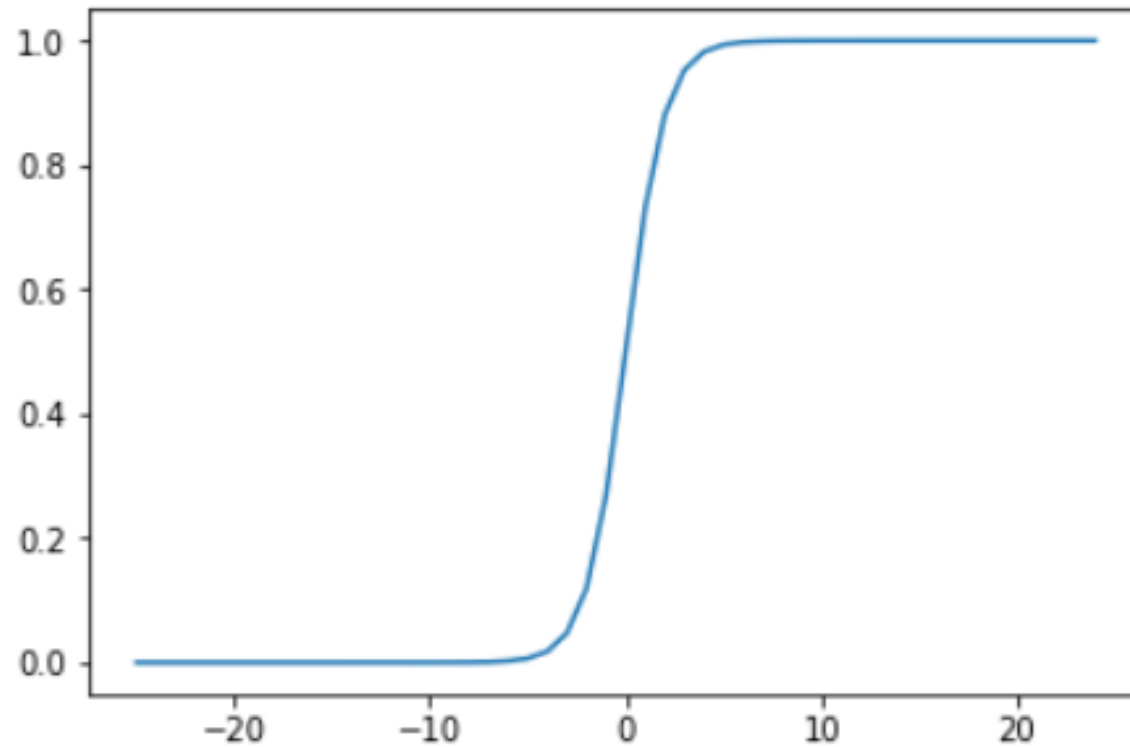
$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

$$z^{(i)} = \mathbf{W} \cdot \mathbf{X} + b$$

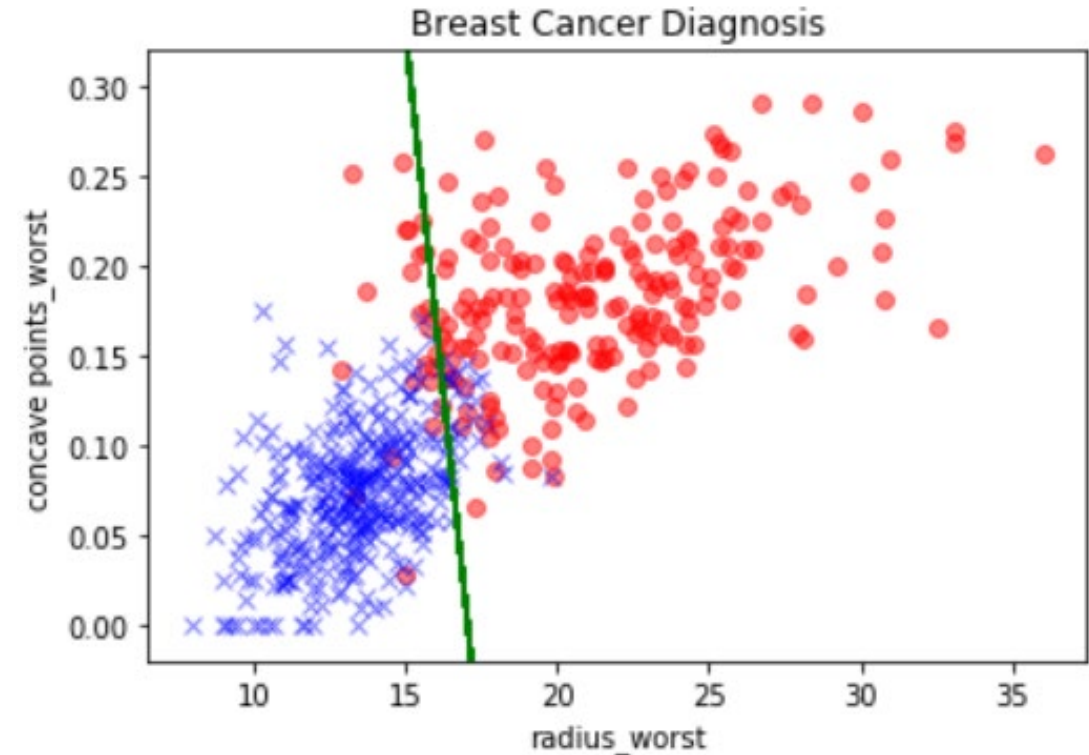
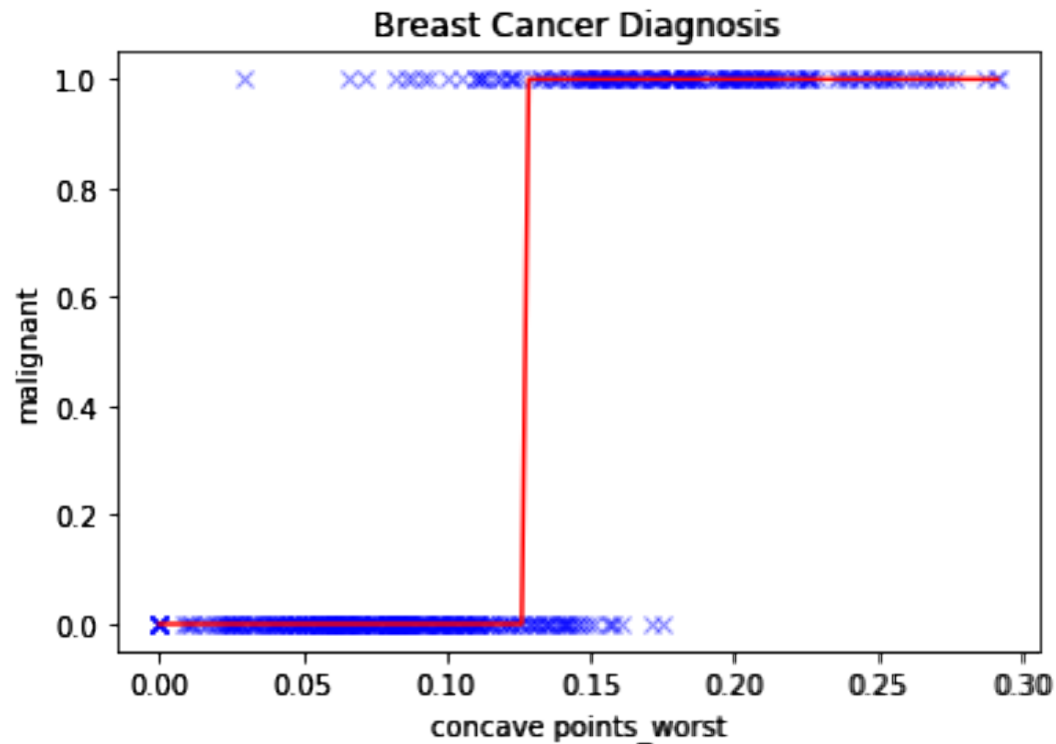


Called "logit" and is related to the decision boundary

$$P^{(i)} \in \mathbb{R}[0, 1]$$

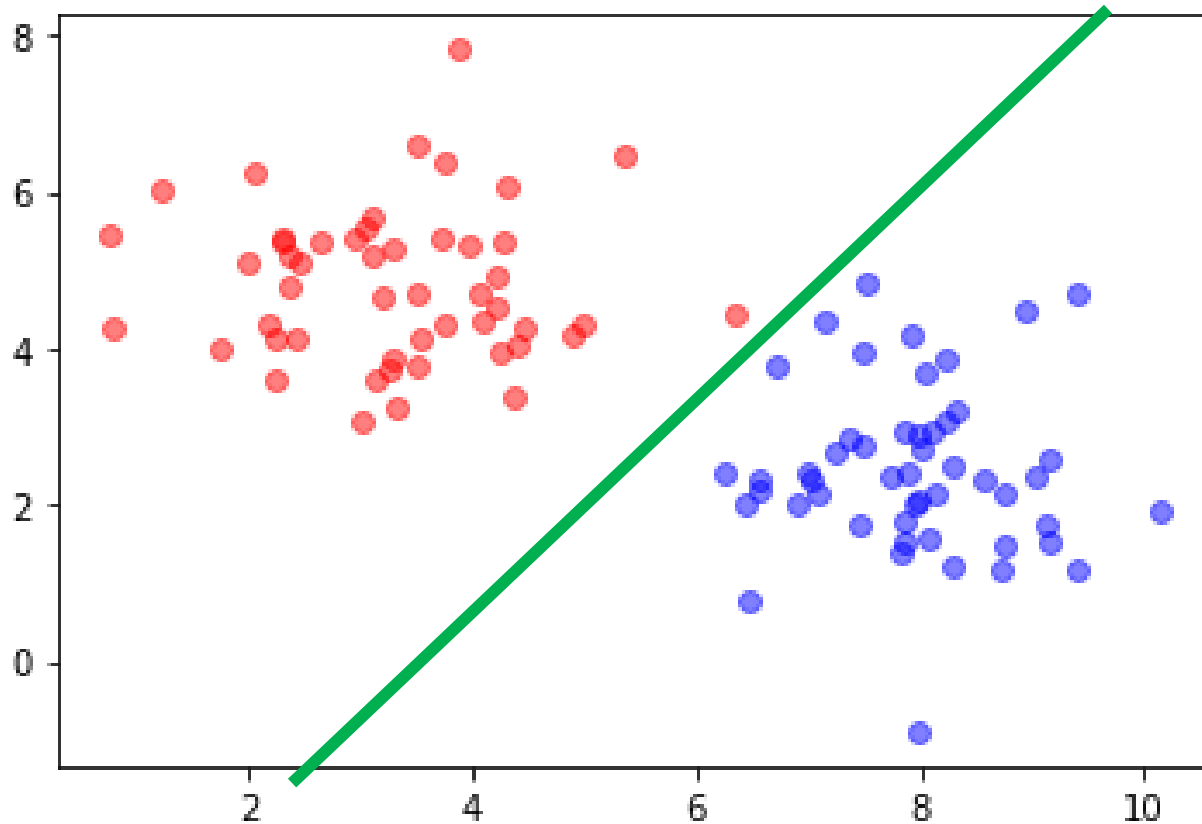


Review: Logistic Regression Decision Boundary



$$z = 0.443 x_1 + 2.76 x_2 - 7.57 = 0$$

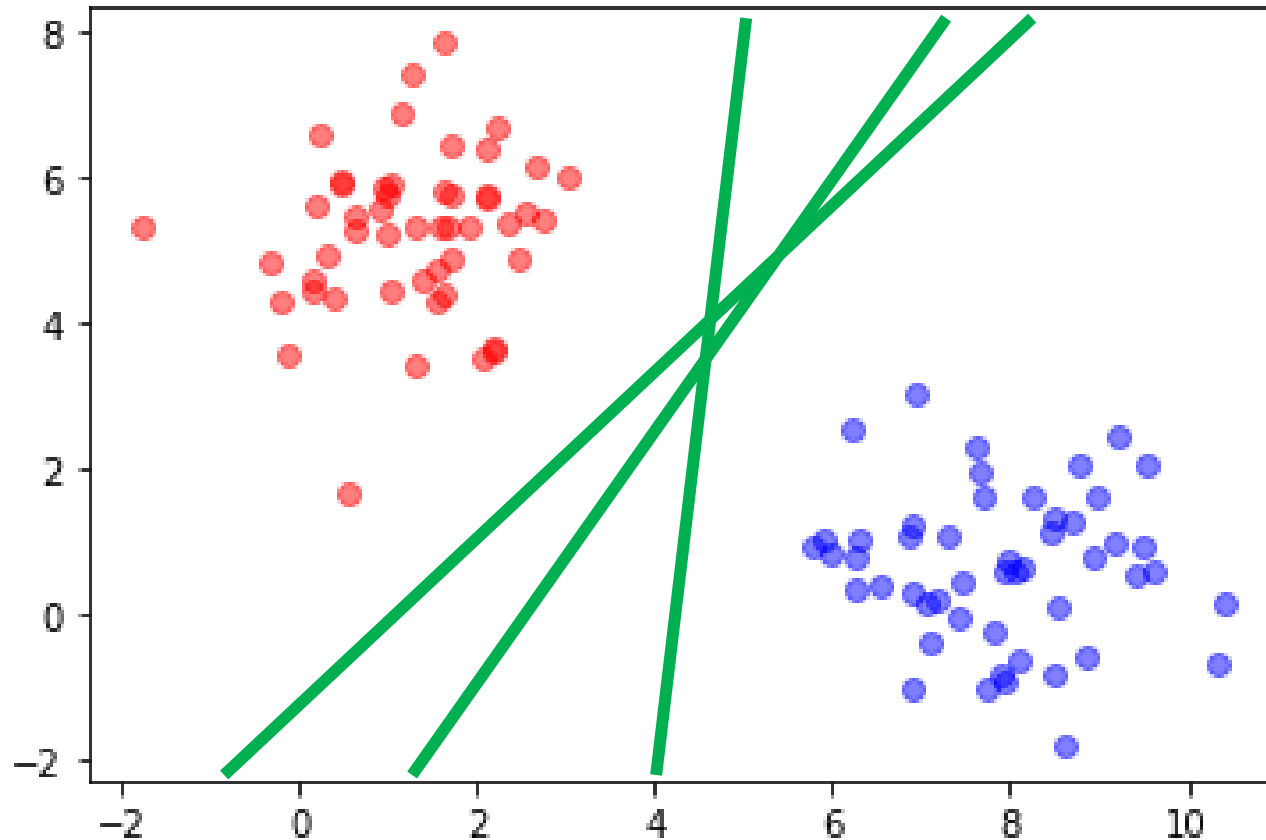
Hyperplane as a Decision Boundary



We can separate the two classes using a hyper plane!

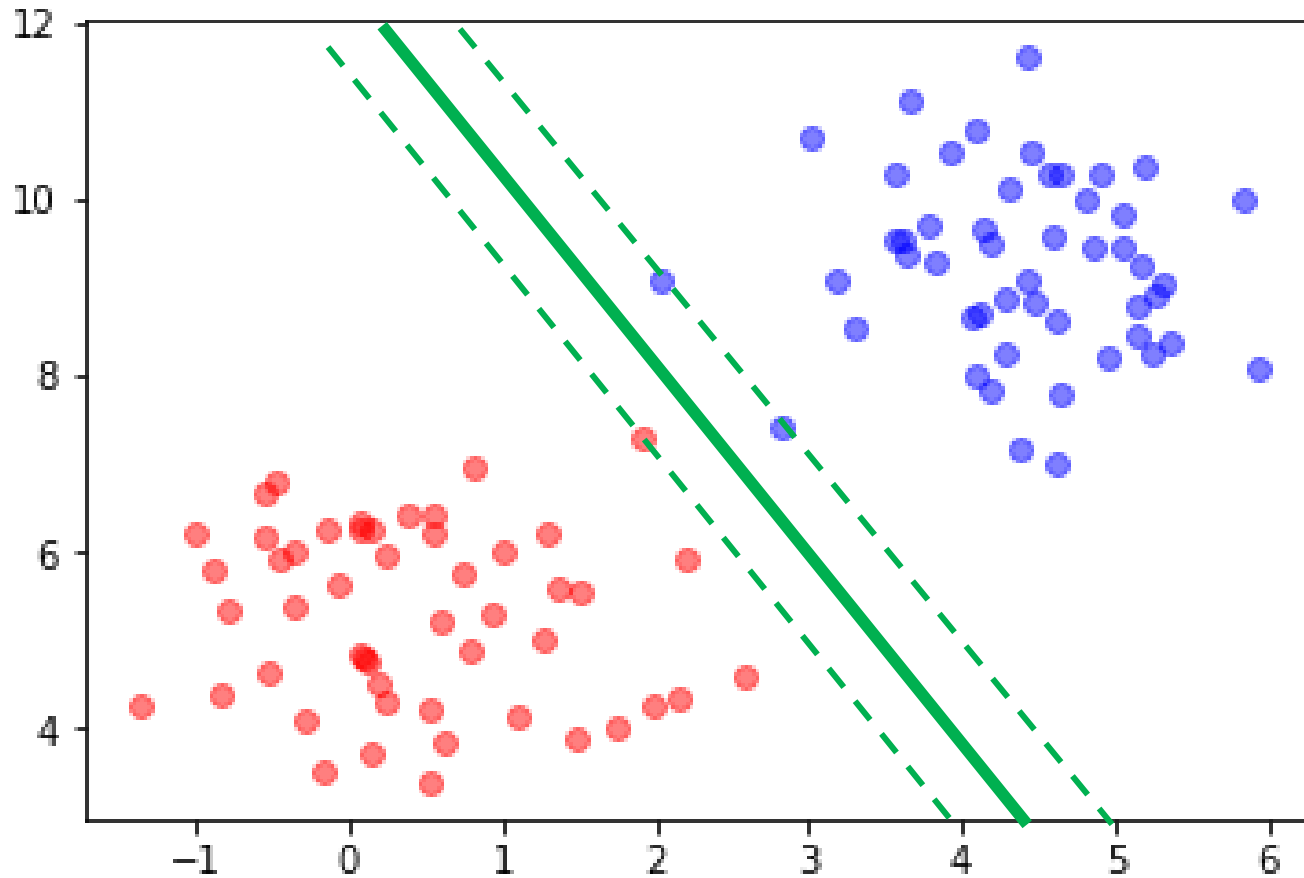
This hyperplane is called “separating hyperplane”

Hyperplane as a Decision Boundary



But which hyperplane should we choose?

Maximum margin classifier

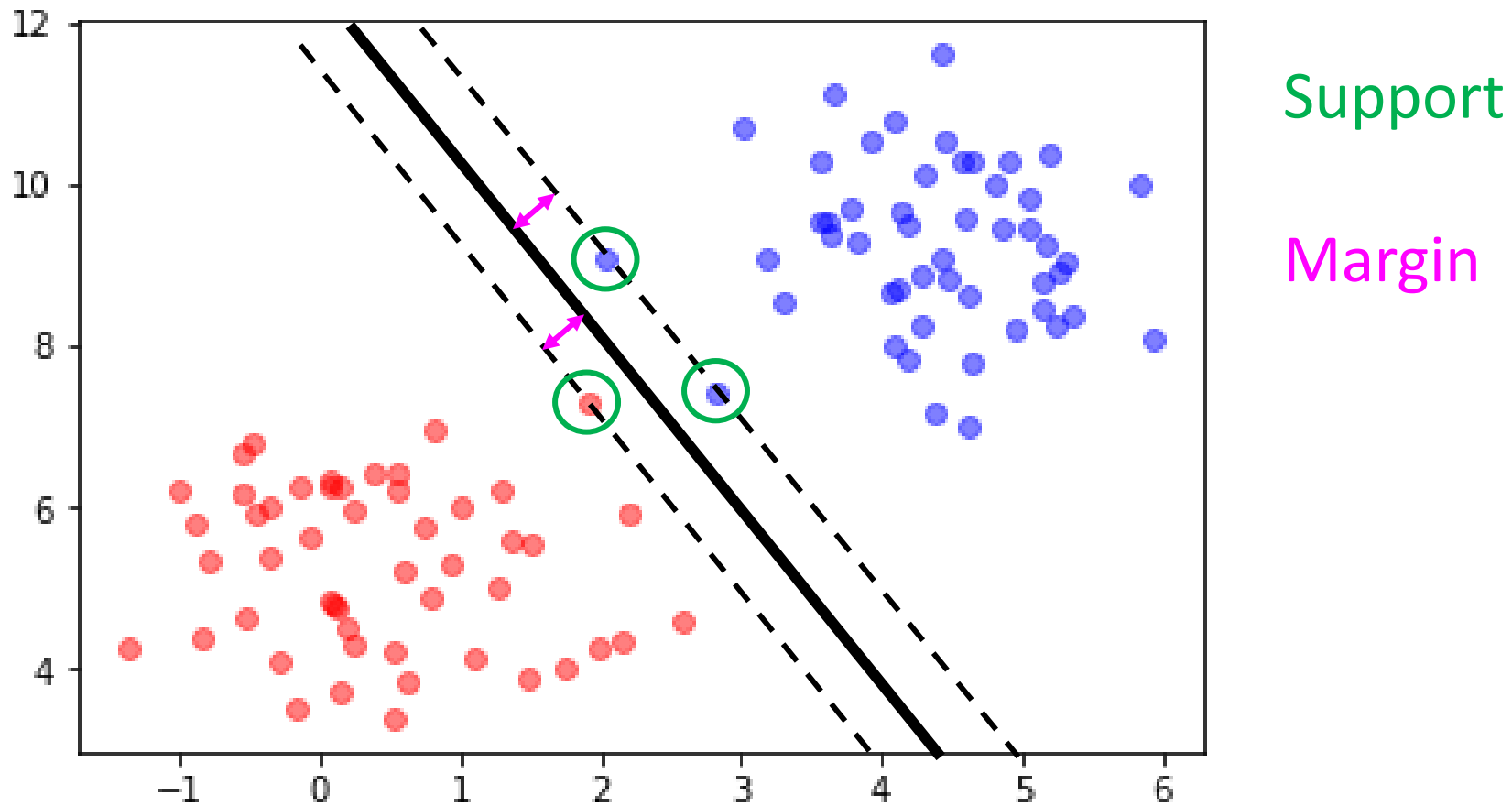


Which hyperplane should we choose?

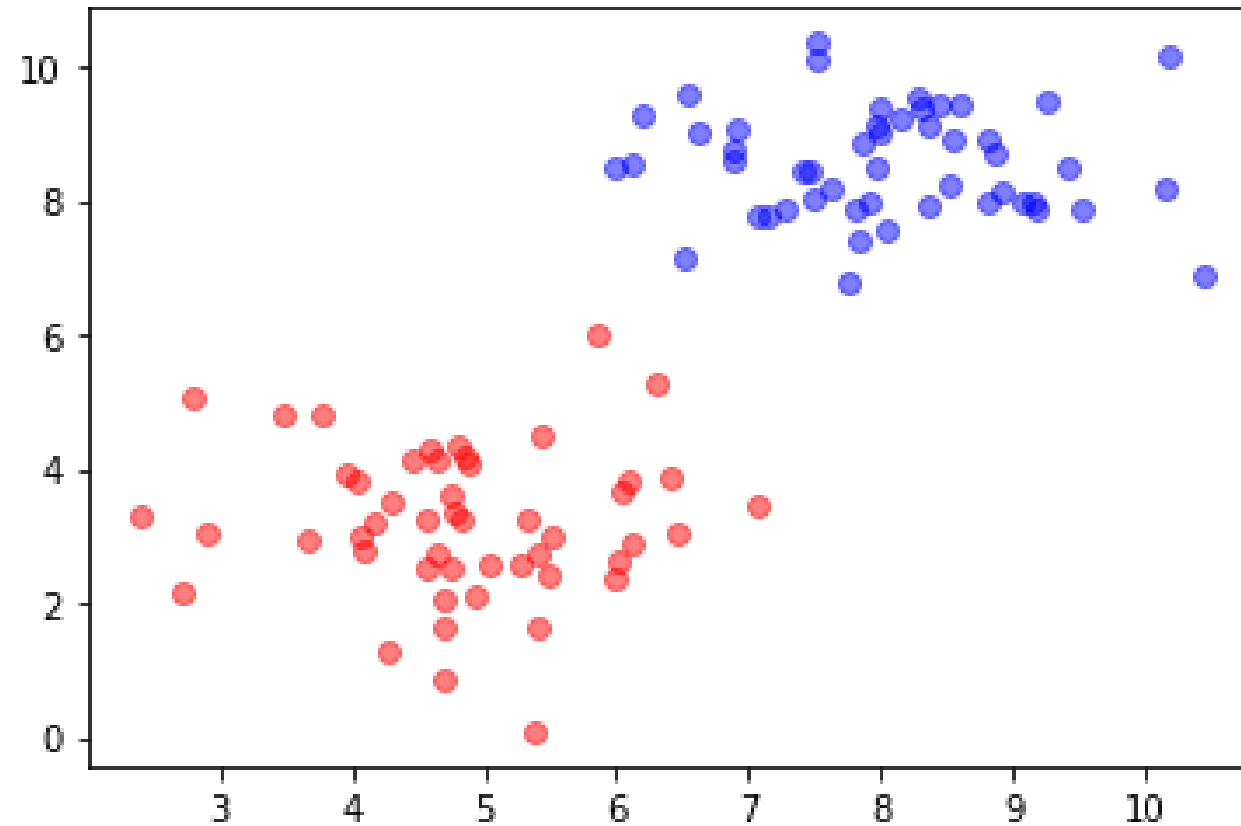
The one with the least likely to misclassify the test data

= The one with the biggest margin

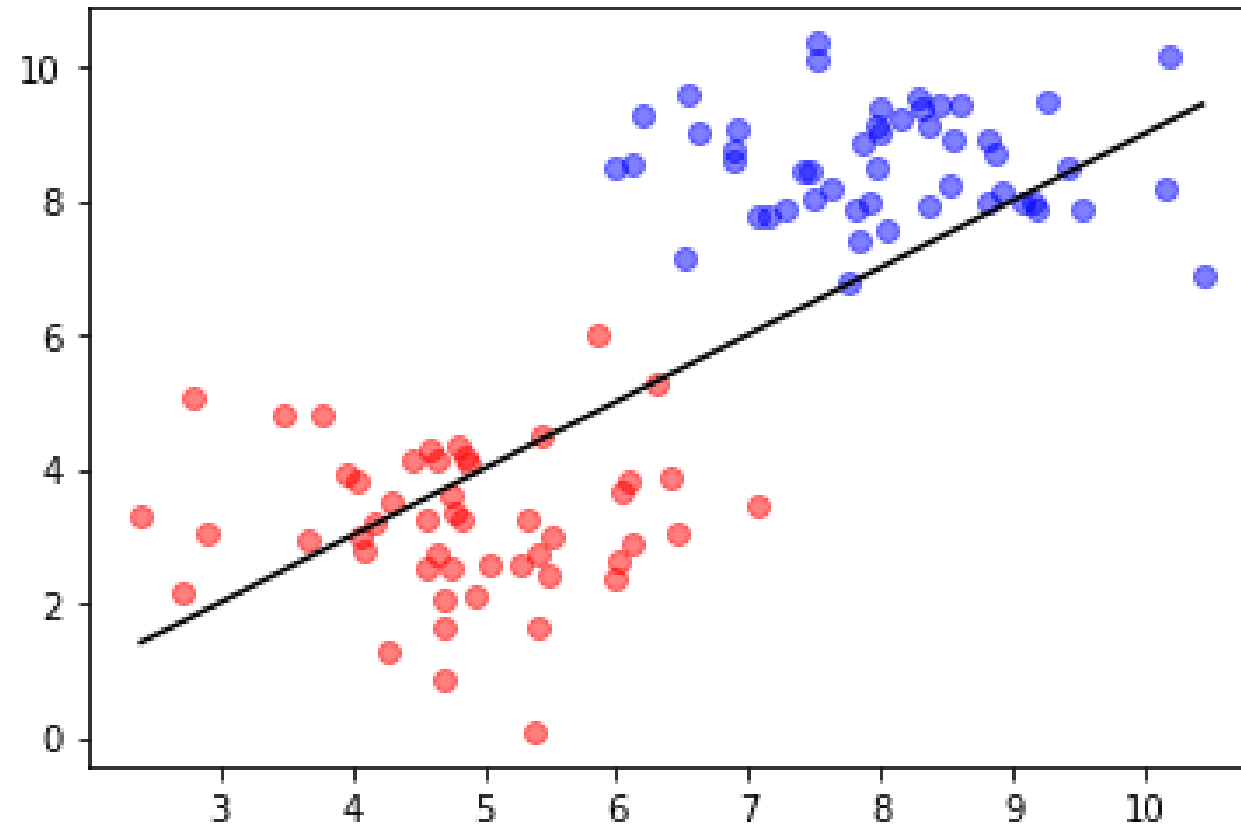
Maximum margin classifier



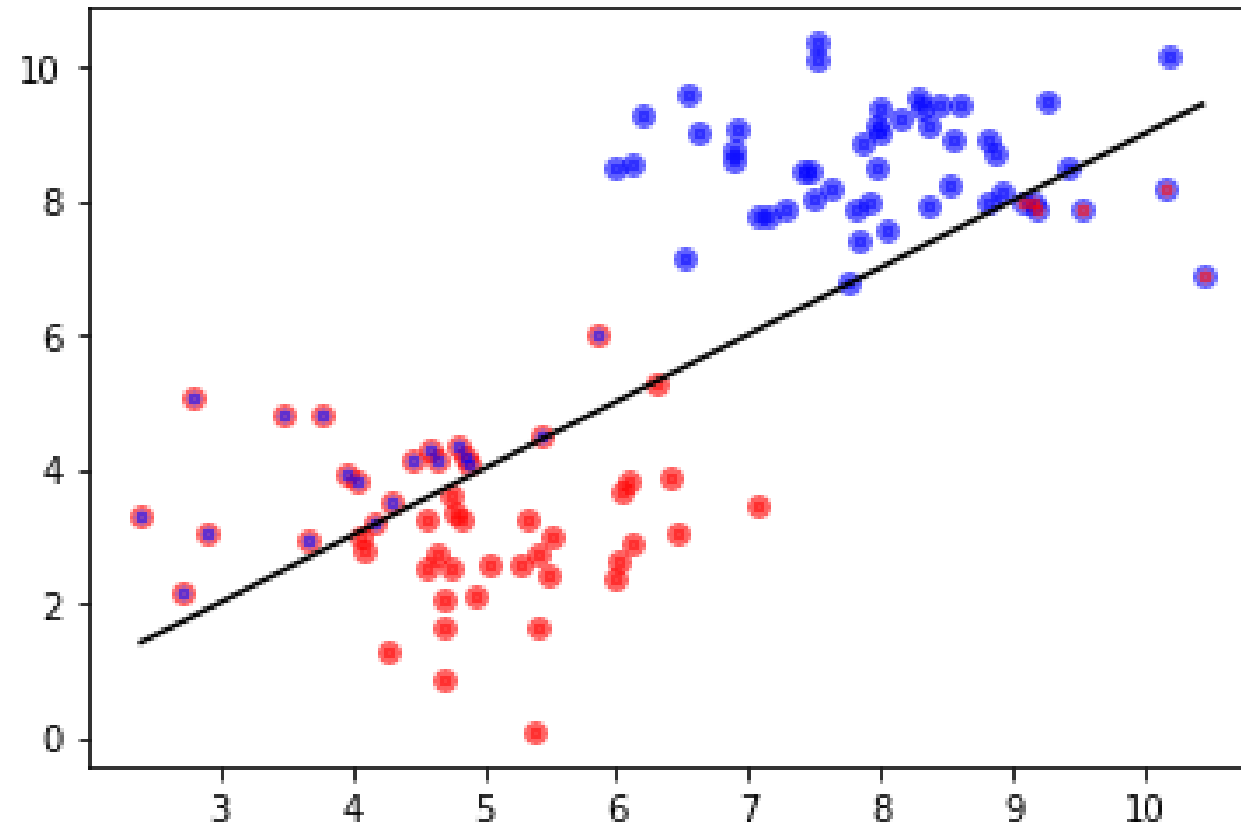
How to find the maximal margin hyperplane?



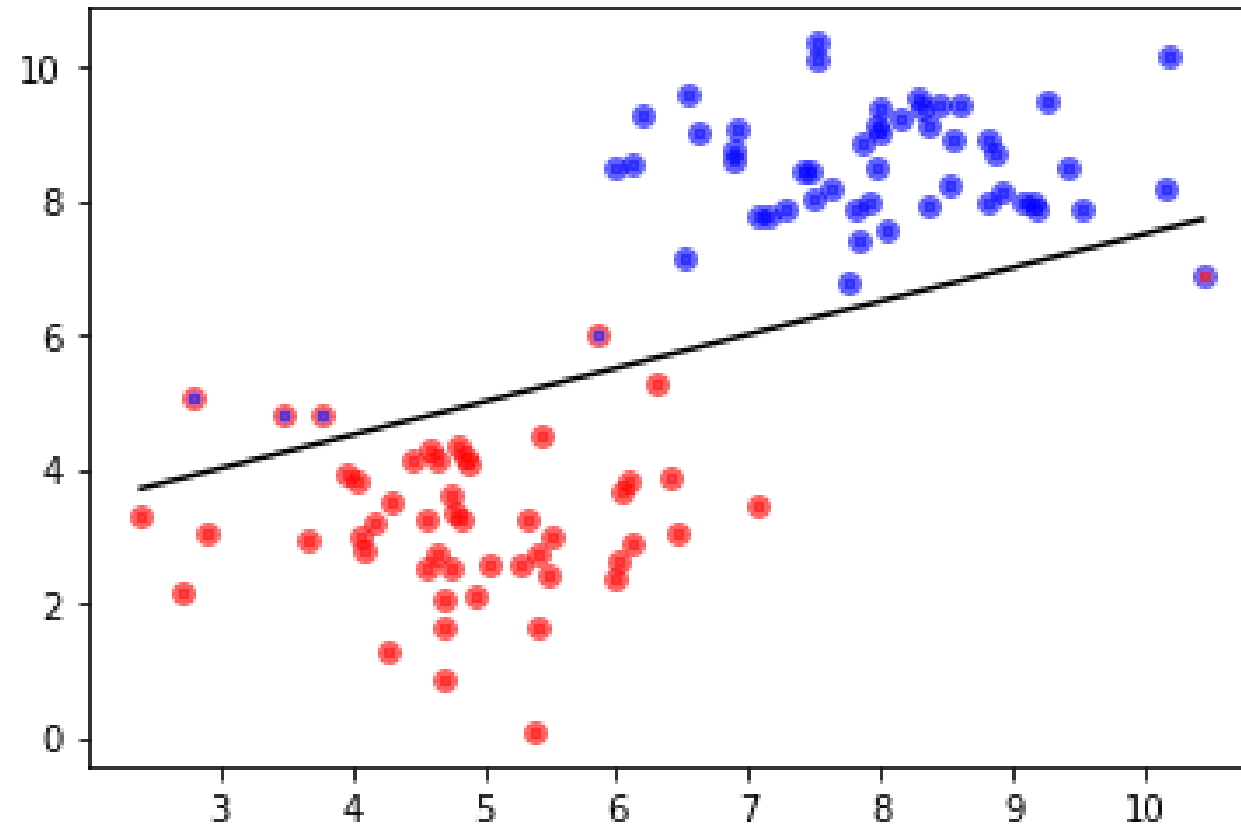
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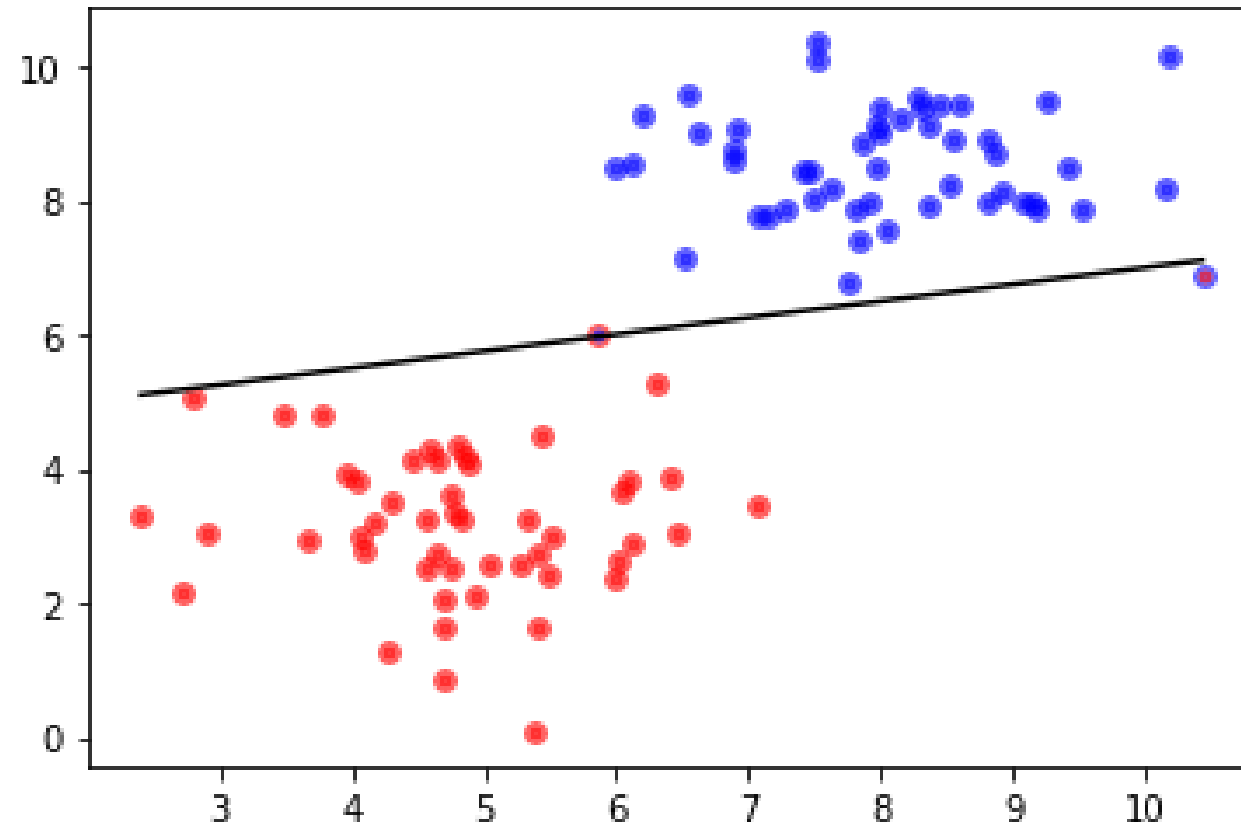
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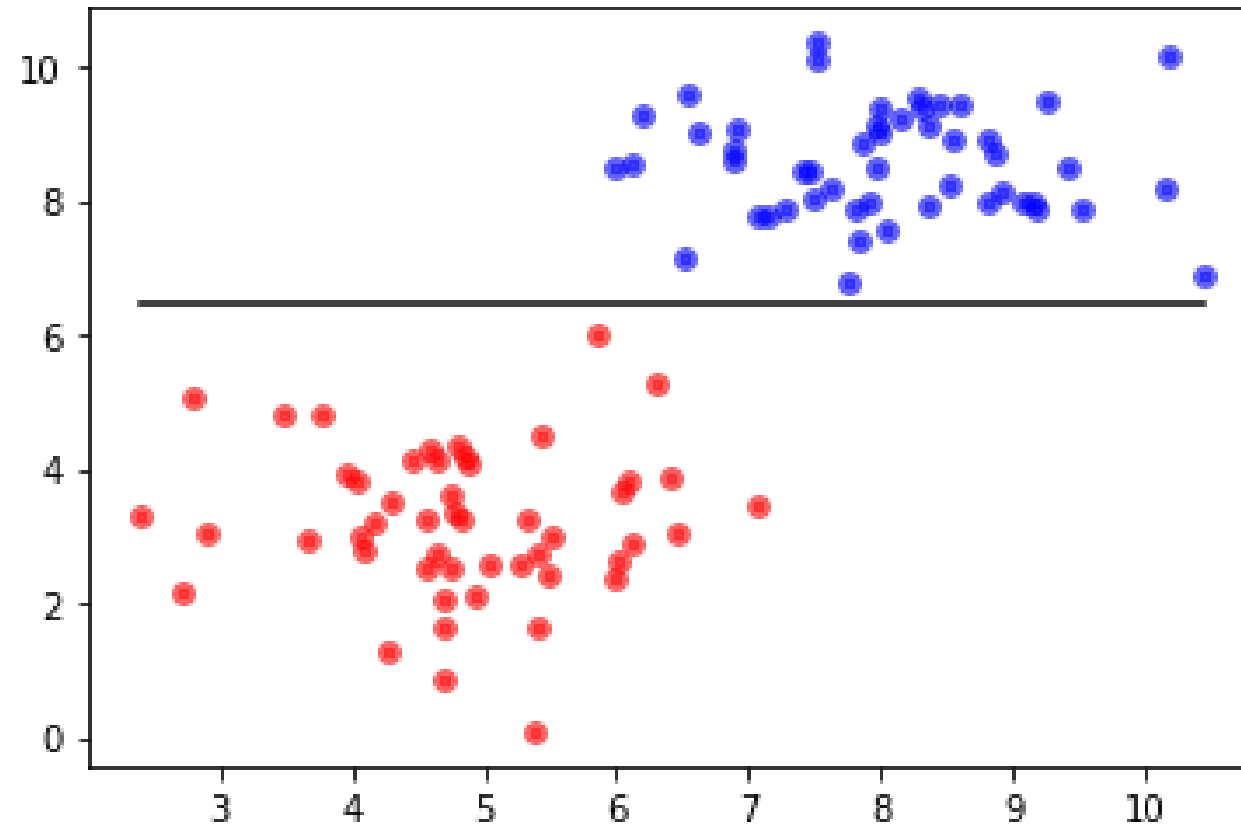
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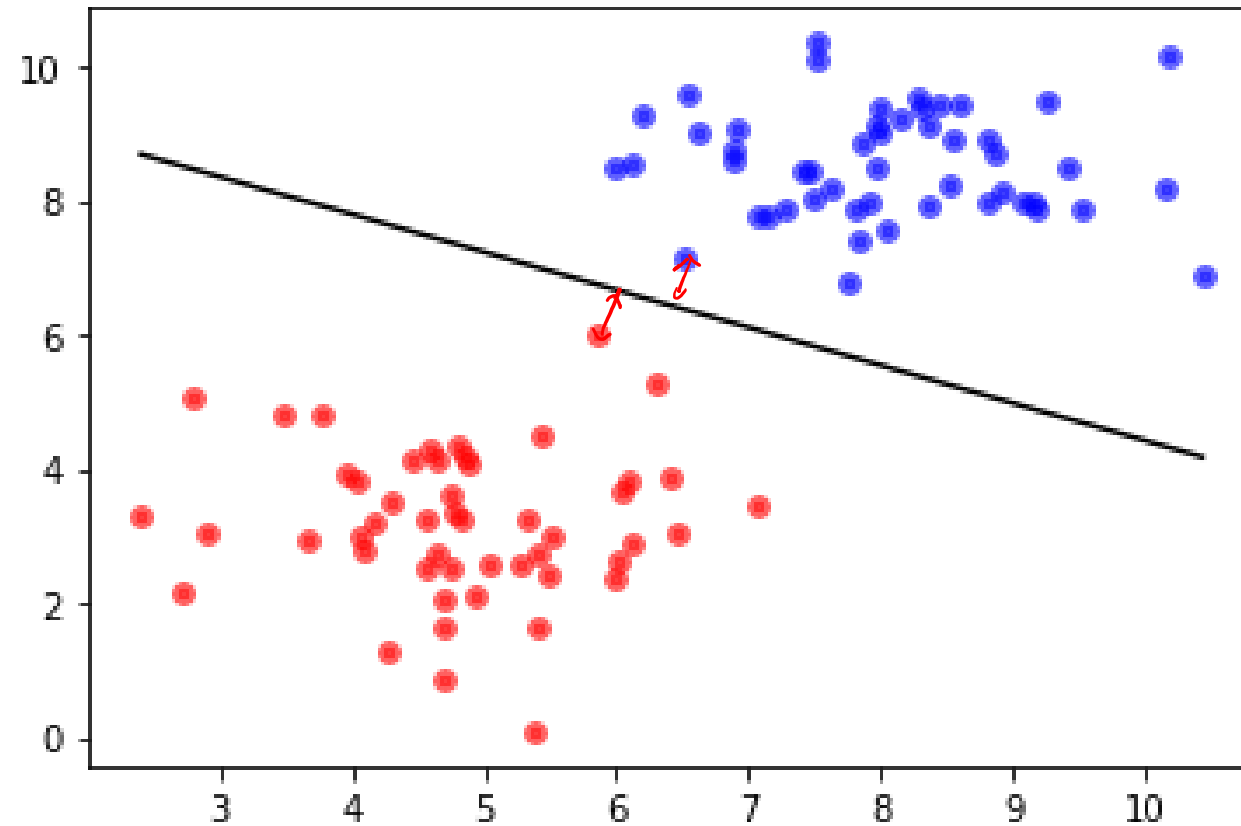
How to find the maximal margin hyperplane?



How to find the maximal margin hyperplane?

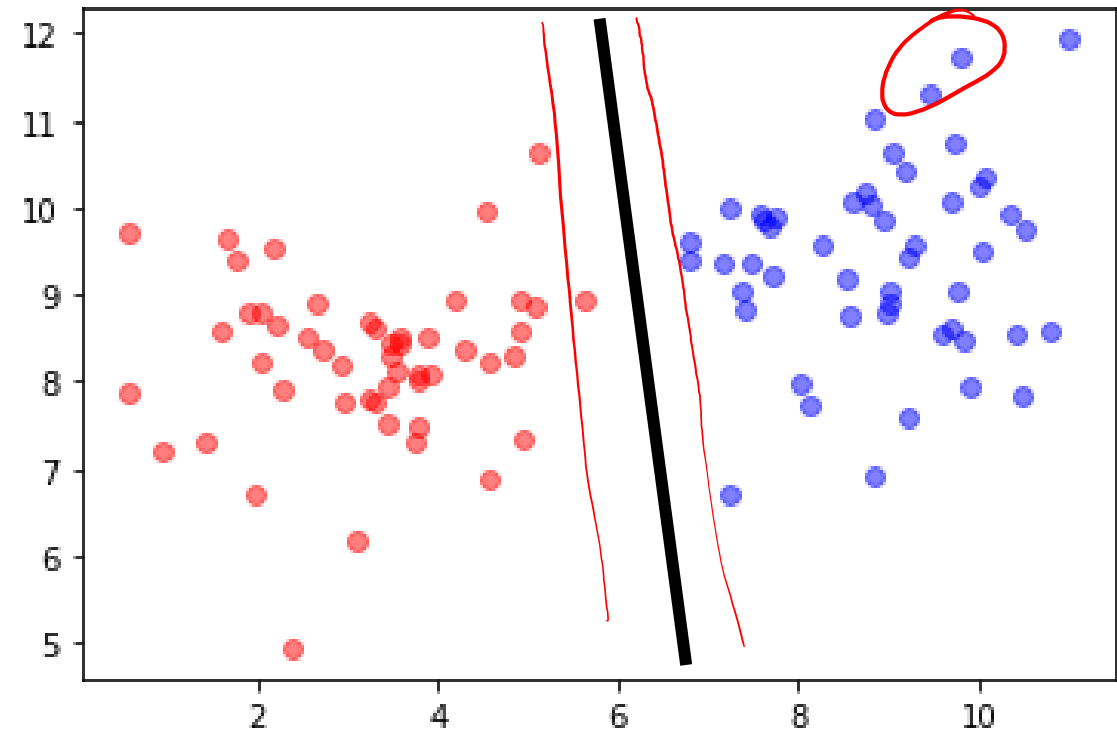
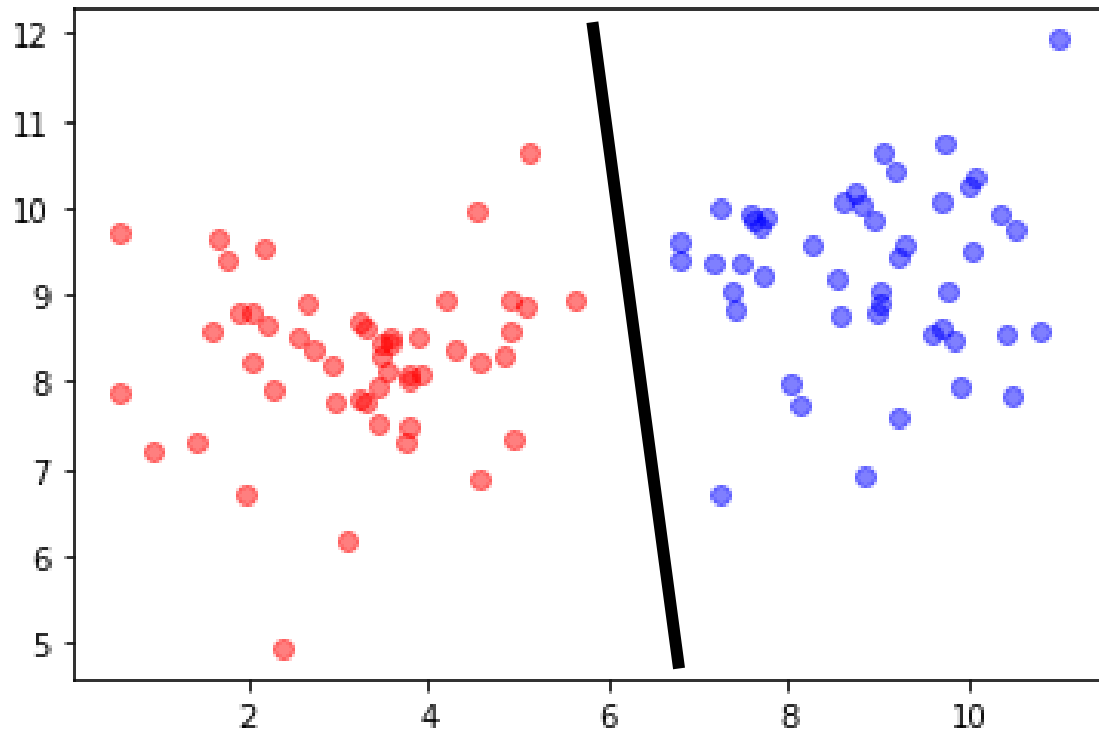


How to find the maximal margin hyperplane?



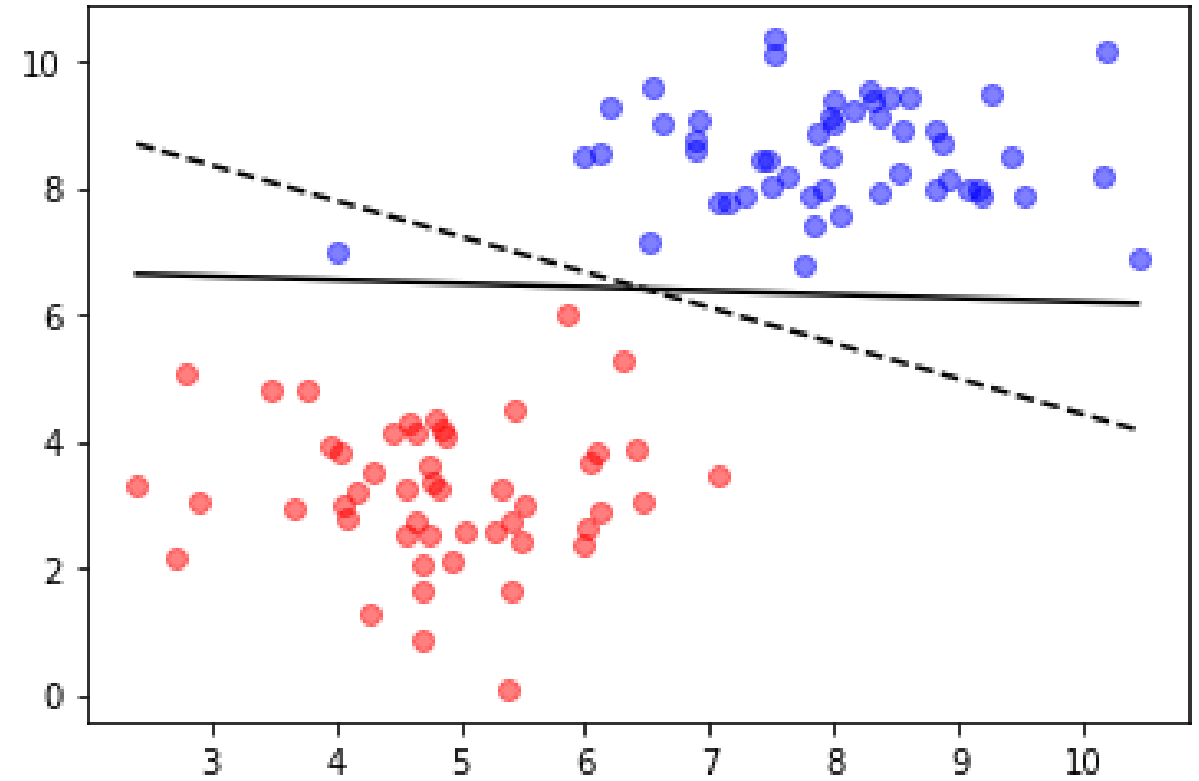
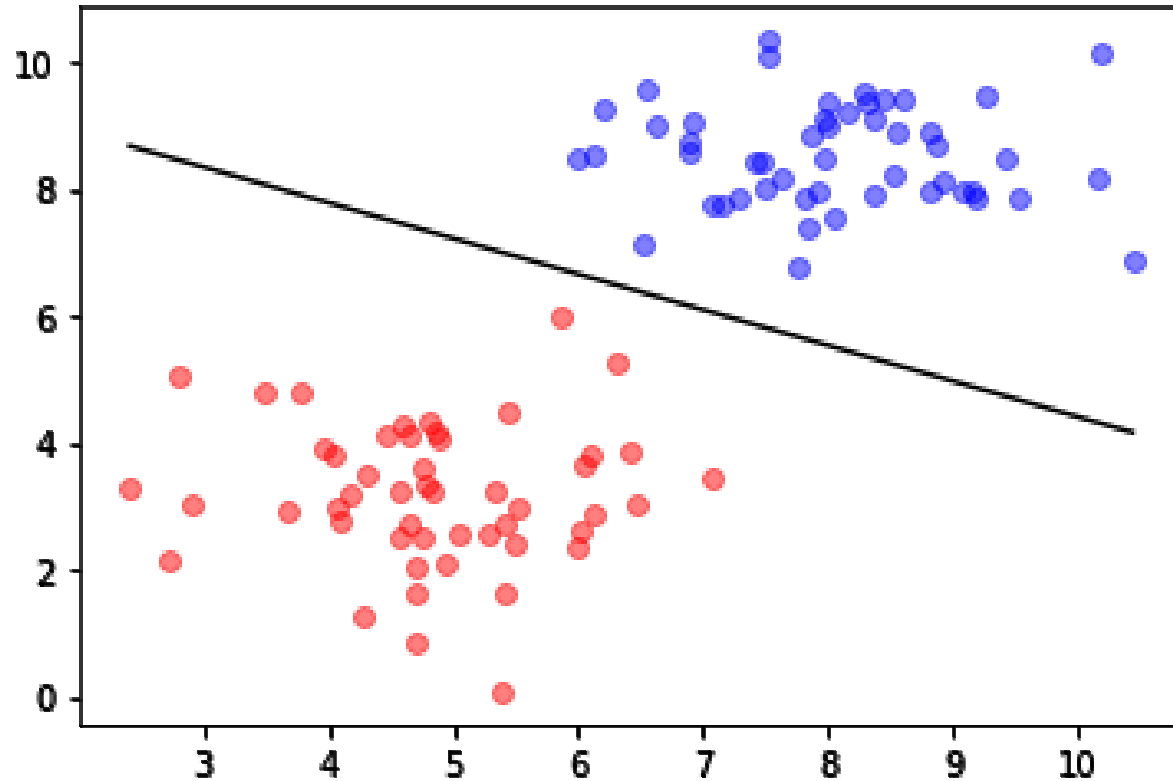
Quiz

What happens to the separating hyperplane when adding new train data points?

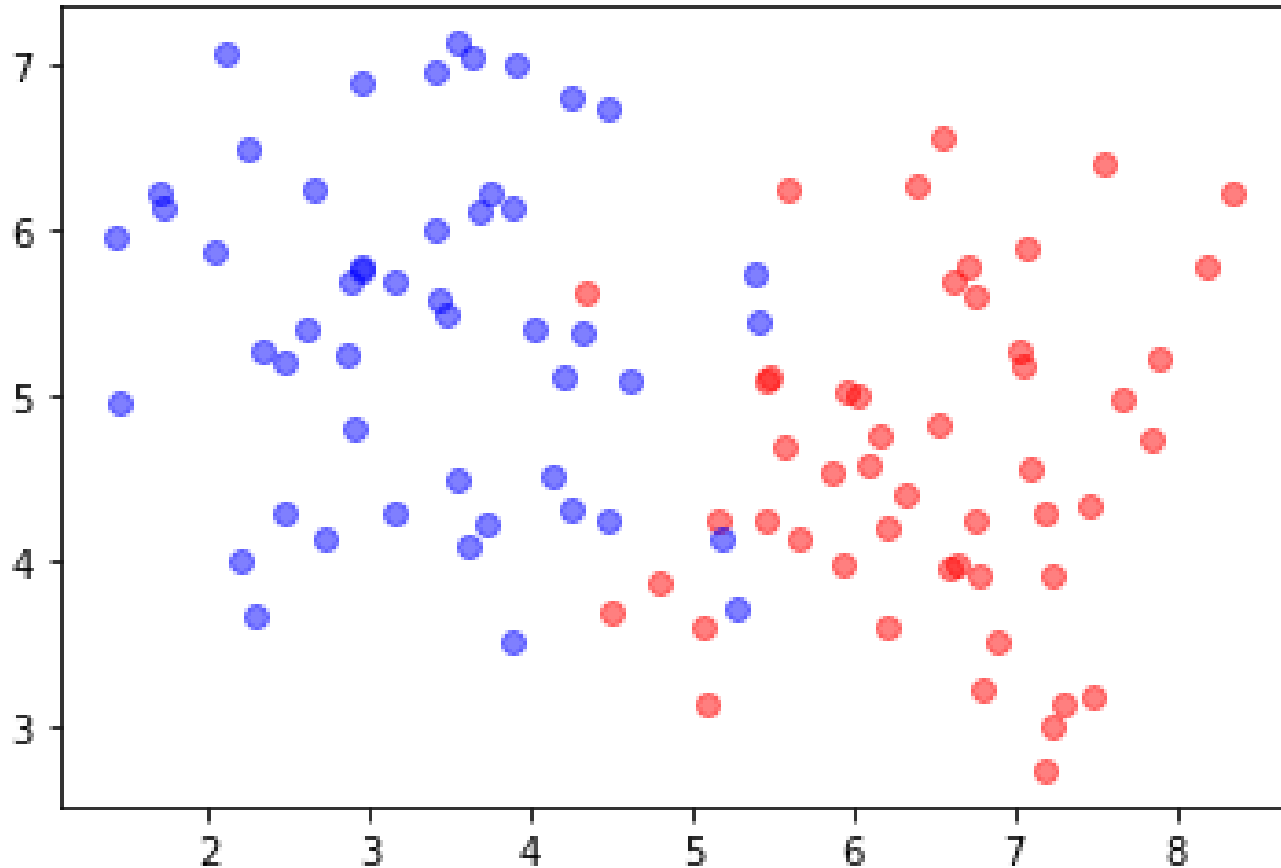


Drawback of the hard margin

What happens to the separating hyperplane?



The impossible case...



Can you separate this with a hyperplane?